The Universe in the Poincare Sphere

0/0 Cosmology: Singularities, Conformal Compactification, and the Removable Value of Spacetime

We show that the fundamental singularities of cosmology -- the Big Bang, spatial infinity, and the Planck scale -- are all 0/0 singularities in the Poincare sphere framework. The universe exists because these singularities are removable. The removable value determines the geometry, the cosmological constant, and the initial conditions.

1. The Poincare Sphere

The Poincare sphere (conformal compactification) maps an unbounded spacetime onto a finite region by rescaling the metric:

g_phys = Omega^2 * g_tilde

where Omega is the conformal factor and g_tilde is the rescaled metric. The boundary of the compactified region corresponds to Omega = 0 (where the conformal factor vanishes).

The Conformal Compactification

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Original metric:     ds^2 = -dt^2 + a(t)^2 * (dx^2 + dy^2 + dz^2)
Conformal time:      d_eta = dt / a(t)
Conformal metric:    ds^2 = a(eta)^2 * (-deta^2 + dx^2 + dy^2 + dz^2)
Rescaled metric:     g_tilde = -deta^2 + dx^2 + dy^2 + dz^2
Physical metric:     g_phys = a(eta)^2 * g_tilde

The conformal factor is Omega(eta) = a(eta).
At the Big Bang: a -> 0, so Omega -> 0.
The metric g_tilde is constant (flat).
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KEY: At the Big Bang, both g_phys -> 0 AND Omega -> 0.
This is a 0/0: g_phys / Omega^2 = g_tilde = FINITE.
The Big Bang singularity is a REMOVABLE singularity.

2. Friedmann Equation as 0/0

The Friedmann equation governs the expansion of the universe:

H^2 = (8piG/3) * rho - k/a^2 + Lambda/3

where H = (da/dt)/a is the Hubble parameter, rho is the energy density, k is curvature, and Lambda is the cosmological constant.

Matter Domination

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rho = rho_0 / a^3
H^2 = (8piG*rho_0/3) * a^{-3}
H ~ a^{-3/2}

dt = da / (a*H) ~ a^{1/2} da
t ~ (2/3) * a^{3/2}
a(t) ~ (3t/2)^{2/3}

At t = 0: a = 0 (Big Bang)
Removable value: a/t^{2/3} = (3/2)^{2/3} = 1.3104
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  0 / 0  SINGULARITY
    Numerator: a(t) = 0 (scale factor vanishes) -> 0
    Denominator: t = 0 (time vanishes) -> 0
    Removable value: a/t^{2/3} = (3/2)^{2/3} = 1.3104
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Radiation Domination

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rho = rho_0 / a^4
H^2 = (8piG*rho_0/3) * a^{-4}
H ~ a^{-2}
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dt = da / (a*H) = a da
t ~ a^2 / 2
a(t) ~ (2t)^{1/2}

At t = 0: a = 0 (Big Bang)
Removable value: a/t^{1/2} = sqrt(2) = 1.4142
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0 / 0 SINGULARITY

```
Numerator: a(t) = 0 (scale factor vanishes) -> 0
Denominator: t = 0 (time vanishes) -> 0
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`Removable value: a/t^{1/2} = sqrt(2) = 1.4142`

Lambda Domination (de Sitter)

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rho = Lambda / (8piG) = const
H^2 = Lambda/3 = const
a(t) = exp(H*t) with H = sqrt(Lambda/3)

At t = 0: a = 1 (NOT zero!)
=> NO Big Bang in pure de Sitter space!
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KEY: The cosmological constant Lambda RESOLVES the Big Bang singularity. In pure de Sitter (Lambda > 0, no matter), there is no Big Bang -- the universe has always existed. Lambda is the 'removable value' that heals the 0/0.

3. de Sitter Space: Conformal Compactification

de Sitter space is the maximally symmetric solution with Lambda > 0.
In conformal coordinates, the metric becomes conformally flat:

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ds^2 = (1/(H*eta)^2) * (-deta^2 + dx^2 + dy^2 + dz^2)

where eta = -exp(-Ht)/H is the conformal time, eta in (-1/H, 0).
Scale factor: a(eta) = 1/(H*|eta|)
Conformal factor: Omega(eta) = 1/(H*|eta|)
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0/0 at the Conformal Boundary

The conformal boundary has two limits:

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AS eta -> 0- (past):
  a(eta) -> infinity  (spatial infinity, NOT a Big Bang)
  Omega -> infinity    (conformal factor blows up)
  g_phys = Omega^2 * g_tilde = (1/(H*eta)^2) * 1 = FINITE
  => The boundary is a REMOVABLE singularity

AS eta -> -1/H (future):
  a(eta) = 1           (finite!)
  Omega = 1            (finite!)
  => Future infinity is FINITE in conformal coordinates
  => The entire future of de Sitter fits in a FINITE region
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PENROSE DIAGRAM OF DE SITTER:
The diagram is a DIAMOND (not a triangle as in Minkowski).
Future infinity i+ is TIMELIKE (not spacelike).
This is because Lambda > 0 makes expansion accelerate,
so light rays can reach spatial infinity in finite time.
The diamond is the 0/0 universe: finite, compact, complete.

Penrose Diagram Construction

The Penrose diagram is obtained by:
1. Conformal compactification: chi -> chi/(1 + H*|chi|)
   This maps the infinite range chi in (-1/H, 1/H) to (-1, 1).
2. The boundary chi = +/-1 is now at FINITE conformal distance.
3. Light rays travel at 45 degrees (conformal invariance).
4. The causal structure is READABLE from the diagram.

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Penrose coordinates:  T = arctan(H*(eta + t_star)) + arctan(H*(eta - t_star))
                      X = arctan(H*(eta + t_star)) - arctan(H*(eta - t_star))

where t_star = conformal spatial coordinate.

The diagram bounds:
  - T in (-pi/2, pi/2)
  - |X| <= pi/2 - |T|  (diamond shape)
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4. The Cosmological Constant as Removable Value

The cosmological constant Lambda appears in Einstein's equations as:

G_ij + Lambda * g_ij = 8piG * T_ij

In the Friedmann equation:
H^2 = (8piG/3) * rho_total
rho_total = rho_matter + rho_radiation + rho_vacuum
rho_vacuum = Lambda / (8piG)

The 0/0 of Vacuum Energy

The vacuum energy density is:

rho_vac = Lambda / (8piG) ~ 10^{-122} in Planck units

This is the most famous 0/0 in physics:
- Quantum field theory predicts: rho_vac ~ 1 (Planck units)
- Observation gives: rho_vac ~ 10^{-122}
- The discrepancy is 122 orders of magnitude

In the 0/0 framework:
- The numerator (quantum fluctuations) -> infinity
- The denominator (gravitational suppression) -> infinity
- The removable value is rho_vac = 10^{-122}
- This is a FINE-TUNED removable value

The cosmological constant problem is a 0/0:
rho_vac = (quantum fluctuations) / (gravitational suppression)
= infinity / infinity
= 10^{-122} (removable value)

The 0/0 framework does NOT solve the fine-tuning problem,
but it IDENTIFIES it as a removable singularity.
The question becomes: what determines the removable value?

5. The Planck Scale: Quantum Gravity 0/0

At the Planck scale (l ~ l_P = sqrt(hbar*G/c^3) ~ 1.6 x 10^{-35} m),
the classical description of spacetime breaks down.

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Wheeler-DeWitt equation:  H |Psi> = 0

This is a 0/0:
  - H = 0 (total energy vanishes)
  - |Psi> = wavefunction of the universe (non-zero)
  - The equation says: the universe has ZERO total energy

Why zero? Because:
  - Gravitational energy is NEGATIVE (bound state)
  - Matter/radiation energy is POSITIVE
  - They cancel EXACTLY: E_grav + E_matter = 0
  - This is the 0/0 of quantum cosmology
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The Wheeler-DeWitt equation H|Psi> = 0 is the ULTIMATE 0/0:

Numerator: |Psi> = path integral over all geometries

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          = int D[g] exp(-S[g]/hbar)
          = FINITE (well-defined)

Denominator: H = 0 (Hamiltonian constraint)

Removable value: |Psi> itself!

The universe is the removable value of its own
partition function. The wavefunction IS the solution
to the 0/0 singularity.
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6. The Complete 0/0 Universe

The universe has four fundamental 0/0 singularities:

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1. BIG BANG:      a -> 0, t -> 0
                  Removable value: a/t^alpha = const
                  (alpha = 2/3 matter, 1/2 radiation)

2. SPATIAL INF:   a -> infinity, t -> infinity
                  Removable value: Lambda/(8piG) = rho_vac
                  (cosmological constant)

3. PLANCK SCALE: l -> l_P, delta_g -> 1
                  Removable value: |Psi> = path integral
                  (Wheeler-DeWitt wavefunction)

4. CONFORMAL BD: Omega -> 0, g_tilde -> infinity
                  Removable value: g_phys = Omega^2 * g_tilde
                  (Penrose diagram boundary)
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What the 0/0 Framework Explains

1. WHY THE BIG BANG IS NOT A TRUE SINGULARITY:
   The conformal factor Omega and the metric g both vanish,
   but their ratio g/Omega^2 = g_tilde is finite.
   The Big Bang is a removable singularity.

2. WHY LAMBDA IS SMALL BUT NOT ZERO:
   Lambda is the removable value of the vacuum energy 0/0.
   It is small (10^{-122}) because of fine-tuning in the
   numerator and denominator.

3. WHY THE WHEELER-DEWITT EQUATION WORKS:
   H|Psi> = 0 is a 0/0. The wavefunction |Psi> is the
   removable value. The universe exists because the
   singularity is removable.

4. WHY THE PENROSE DIAGRAM IS FINITE:
   The conformal compactification maps infinite spacetime
   onto a finite diamond. The boundary is a removable
   singularity (0/0 in the conformal factor).

What the 0/0 Framework Does NOT Explain

1. WHY the removable value has the specific value it does
   (e.g., why Lambda ~ 10^{-122} and not some other value)

2. WHAT happens INSIDE the singularity (before t=0)
   The 0/0 framework says the singularity is removable,
   but does not specify what the universe 'was' before.

3. HOW quantum gravity resolves the Planck scale 0/0
   The Wheeler-DeWitt equation is the candidate, but
   a full solution is not known.

The Poincare Sphere Picture

In the Poincare sphere, the entire history of the universe
is contained in a finite ball:

- The center is the Big Bang (0/0, removable)
- The equator is spatial infinity (0/0, removable)
- The poles are temporal infinity (0/0, removable)
- The surface is the conformal boundary (0/0, removable)

Every boundary of the universe is a 0/0 singularity.
Every 0/0 is removable.
The universe is the removable value of its own boundary.

This is the deepest statement of the 0/0 framework:
THE UNIVERSE EXISTS BECAUSE ALL ITS SINGULARITIES
ARE REMOVABLE.

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